DEPARTMENT OF ECONOMICS
B.Sc. ECONOMICS
SECOND SEMESTER, 2023/2024 EXAMINATION
ECN 332 – RESEARCH METHODOLOGY

Lecturer: Dr. Oyeyemi O. Biyi | **Credit Units: 3**
Instruction: Attempt any four (4) questions | **Time Allowed: 2 Hours**

QUESTION 1

Research is a process of systematic inquiry that entails the collection of data, documentation of critical information, etc., by suitable methodologies set by specific fields or academic disciplines. Explain the classes of research. (15 marks)

(Total: 15 marks)

QUESTION 2

(a) Choose a researchable topic in the field of Economics. (3 marks)
(b) State two research objectives from the topic chosen. (4 marks)
(c) Formulate two hypotheses from your research objectives. (4 marks)
(d) Explain two errors in hypothesis testing. (4 marks)

(Total: 15 marks)

QUESTION 3

(a) Define a variable. (2 marks)
(b) Discuss any four (4) types of variables. (6 marks)
(c) State the differences between a variable and a concept in research. (3 marks)
(d) Briefly discuss any two (2) methods/approaches of analysing data in the field of Economics. (4 marks)

(Total: 15 marks)

QUESTION 4

Tolulope Elohor Obele and Oyeyemi Omodadepo Adebiyi made the statement below in an article titled, 'Quantitative Assessment of the Effect of Government Expenditure on Poverty Incidence in Nigeria' published by the Journal of Business and Economic Analysis written in 2023:

"A reasonable government spending on building and construction, education, health, security, and transportation serve as an engine of economic growth which in turn translate into decreasing the incidence of poverty in Nigeria."

The journal is the fifth volume and number three with DOI: 10.1142/S2737566823500032.

<u>Required:</u>

(a) Use APA 7 referencing style to write the in-text citation. (5 marks)
(b) Use APA 7 referencing style to write the reference. (6 marks)
(c) What is plagiarism? (4 marks)

(Total: 15 marks)

QUESTION 5

(a) Explain any four (4) sampling techniques. (6 marks)
(b) Outline any six (6) differences between quantitative and qualitative research (9 marks)

(Total: 15 marks)

QUESTION 6

(a) Explain three (3) ways in which literature review can help your study. (6 marks)
(b) Discuss any two (2) sources where you can get existing literature. (4 marks)
(c) State the stages involved in research process. (5 marks)

(Total: 15 marks)

ECN 322 – APPLIED STATISTICS II

Lecturer: Dr. Oyeyemi O. Ilyi | Credit Units: 3

Instruction: Attempt any four (4) questions | Time Allowed: 2 Hours

QUESTION 1

The breakdown of the student body in a local high school according to race and ethnicity is 51% white, 27% black, 11% Hispanic, 6% Asian, and 5% for all others. A student is randomly selected from this high school. Find the probabilities of the following events:

(a) The student is black. (4 marks)

(b) The student is minority (5 marks)

(c) The student is not black. (6 marks)

(Total: 15 marks)

QUESTION 2

Two fair dice are thrown.

Required:

(a) Construct a table containing the possible outcomes. (3 marks)

(b) Find the probability of the event that both dice show a four. (5 marks)

(c) Find the probability of the event that at least one die shows a four. (7 marks)

(Total: 15 marks)

QUESTION 3

A service organisation in a large town organises a raffle each month. One thousand raffle tickets are sold for $1 each. Each has an equal chance of winning. First prize is $300, second prize is $200, and third prize is $100. Let X denote the net gain from the purchase of one ticket.

Required:

(a) Construct the probability distribution of X. (5 marks)

(b) Find the probability of winning any money in the purchase of one ticket. (5 marks)

(c) Find the expected value of X, and interpret its meaning. (5 marks)

(Total: 15 marks)

QUESTION 4

It is generally believed that children brought up in stable families tend to do well in school. To verify such a belief, a social scientist examined 290 randomly selected students' records in a public high school and noted each student's family structure and academic status four years after entering high school. The data were then sorted into a 2×3 contingency table with two factors. The results are given in the table below.

		Academic Status	
		Graduate	Did not Graduate
Family	No Parent	18	31
	One Parent	101	44
	Two Parents	70	26

Required:

Test, at the 1% level of significance, whether there is sufficient evidence in the data to conclude that family structure matters in school performance of the students. (Note, $\chi^2_{0.01} = 9.210$).

(15 marks)

(Total: 15 marks)

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B.Sc. ECONOMICS
SECOND SEMESTER, 2023/2024 EXAMINATION
ECN 342 – INTRODUCTORY ECONOMETRICS

Lecturer: Mr. Eseyin, Oluwasegun S. **Credit Units: 3**
Instruction: Answer any four (4) questions **Time Allowed: 2 Hours**

QUESTION 1

(a) Explain the meaning of econometrics. **(2 marks)**
(b) Why do we have econometrics as a separate discipline? **(3 marks)**
(c) With relevant examples, discuss in details the methodologies of econometrics. **(5 marks)**
(d) Use relevant examples to discuss dynamic models. **(5 marks)**
(Total: 15 marks)

QUESTION 2

(a) Explain the various assumptions of the Classical Linear Regression Model. **(5 marks)**
(b) Discuss the meaning of heteroskedasticity, outlining how it constitutes a violation of the assumptions of the classical linear regression model, how to detect its presence in a model and ways by which it can be corrected. **(10 marks)**
(Total: 15 marks)

QUESTION 3

Given the following research titles:—

(i) Impact of human capital development on economic growth
(ii) The role of fiscal policy in job creation in Nigeria
(iii) Determinants of household savings in Ede metropolis of Osun State, Nigeria

Required

(a) State the hypothesis as well as theoretical underpinning for each of the research titles and formulate models for analysis (by clearly stating the dependent and the independent variables) for each of them. **(3 marks)**
(b) Justify the rationale for the inclusion of each of the variables into the models. **(4 marks)**
(c) Suggest the forms of data that will be suitable for analysis. **(3 marks)**
(d) What are the likely interpretations for the parameters identified in 3(a) above? **(5 marks)**
(Total: 15 marks)

QUESTION 4

(a) Discuss the meaning of multicollinearity, enumerating how it constitutes a violation of the assumptions of the classical linear regression model, how to detect its presence in a model, and the ways by which it can be corrected. **(7.5 marks)**
(b) Discuss the meaning of serial correlation, stating how it constitutes a violation of the assumptions of the classical linear regression model, how to detect its presence in a model and the ways by which it can be corrected. **(7.5 marks)**
(Total: 15 marks)

QUESTION 5

(a) Given that $Y_{n\times 1} = X_{n\times k} B_{k\times 1} + U_{n\times 1}$, show that $\hat{\beta} = (X'X)^{-1}X'Y$ (5 marks)

(b) Assuming that the demand (y) for a commodity and the corresponding price (x) are given as follows:

Price (x)	45	40	35	32	30	25	24	23	22	20
Demand (y)	500	550	560	580	585	600	610	620	650	670

Required

(i) Compute the linear regression line using the matrix method $\hat{\beta} = (X'X)^{-1}X'Y$ (8 marks)

(ii) What will be the demand if the price is 60 and 15? (2 marks)

(Total: 15 marks)

QUESTION 6

(a) Explain the properties of Gauss-Markov Theorem. (3 marks)

(b) Use the matrix method to prove that the ordinary least squares (OLS) estimator is BLUE. (12 marks)

(Total: 15 marks)

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B.Sc. ECONOMICS
SECOND SEMESTER, 2023/2024 EXAMINATION
ECN 302 – INTERMEDIATE MACROECONOMICS

Lecturer: Olufunmilayo O. Jemiluyi (Ph.D.) | **Credit Unit: 3 Units**
Instruction: Answer any four (4) questions | **Time Allowed: 2 Hours**

QUESTION 1

(a) Explain why the estimates of national income using the three accounting approaches would yield the same results in the absence of estimation error. **(5 marks)**

(b)

Table 1: Hypothetical national income data for country A

Items	Amount (₦' billion)
Income from self employment	38
Public sector profit	5.8
Private sector profit	11.2
Income from employment	24
Consumption of fixed capital	0.16
Interest	4.01
Company tax	2.25
Rent	6.5
Withholding tax (WHT)	0.75
Stock appreciation	0.95
Income of non-nationals resident in country A	29.05
Income of ountry A nationals from abroad	43.5
Other income	4.65
Residual error	1.02

Use the information presented in Table 1 above to answer the following questions:

(a) What approach would you recommend for the estimation of country A's national income?

(b) Using the recommended approach, calculate;

i. Gross Domestic Product (GDP) at factor cost

ii. Gross National Product (GNP) at factor cost

iii. Net National Product (NNP) at factor cost

(10 marks)
(Total: 15 marks)

QUESTION 2

(a) Distinguish between the tenets of the neo-classical and keynesian macroeconomics. **(5 marks)**

(b) Mention the arguments that arose in criticism of Keynes absolute income hypothesis, and discuss in details any one of the arguments. **(10 marks)**

(Total: 15 marks)

QUESTION 3

(a) Given the following statistics for a three-sector economy: Coucmption function $C = C_0 + C_y Y$, Investment $I = I_0$ and Government expenditure $G = G_0$, show that;

i. The investment and government expenditure multipliers are the same